

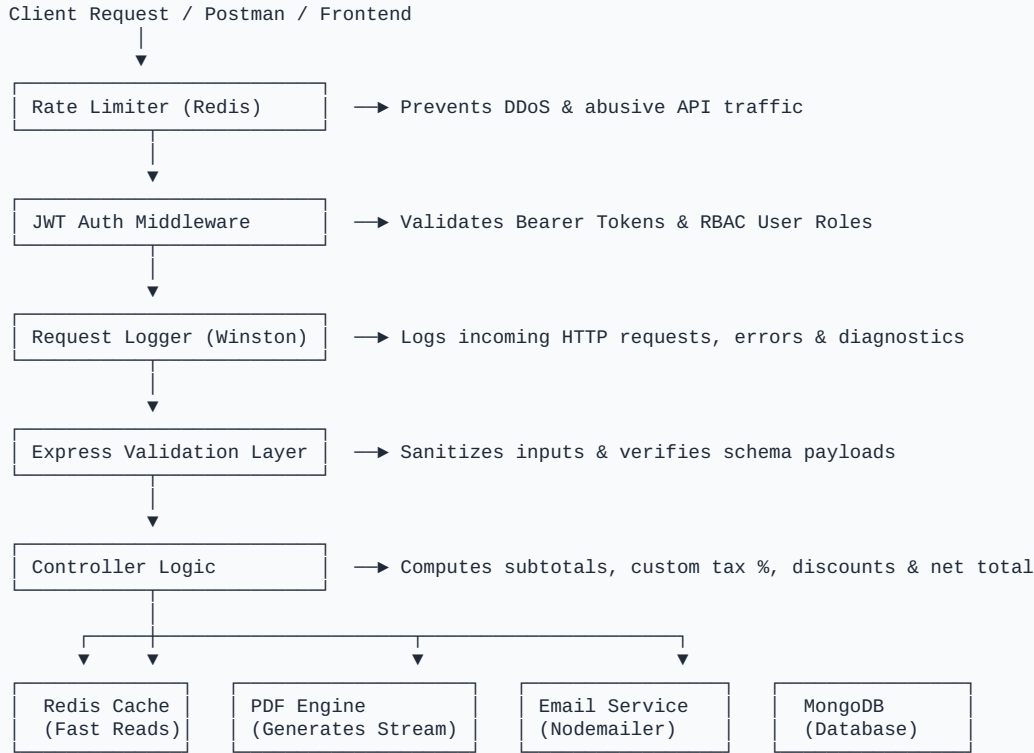


Enterprise Invoice Generator & PDF Engine

Production-ready REST API & automated PDF rendering pipeline with JWT, Redis, Nodemailer, Rate Limiting & i18n.



SYSTEM ARCHITECTURE & PIPELINE



KEY FEATURES & ENHANCEMENTS

- **JWT Auth & Security:** Registration, login, password hashing (bcrypt), and Role-Based Access Control (RBAC).
- **PDF Engine & Downloads:** Dynamic PDF generation streaming memory buffers for instant browser downloading.
- **Nodemailer Dispatch:** Automatic email attachment of generated PDF invoices to client email addresses.
- **Redis Caching & Rate Limiting:** High-performance query caching and rate-limiting to prevent API abuse.
- **i18n Translation & Localization:** Multi-language document strings and dynamic localized currency formatting.
- **Logging & Monitoring:** Centralized Winston logger auditing, error tracking, and PM2 cluster process monitoring.



TECH STACK & DEPENDENCIES

Node.js (v18+) Express.js MongoDB & Mongoose Redis Server JSON Web Tokens (JWT) Bcrypt.js Nodemailer

express-rate-limit Winston & Morgan i18n PM2



LOCAL SETUP & CONFIGURATION

```
# 1. Clone & Install
git clone https://github.com/nikhilsingh2764/invoice-Genrator.git
cd invoice-Genrator && npm install

# 2. Environment Variables (.env)
PORT=5000 | NODE_ENV=development | MONGO_URI=mongodb://127.0.0.1:27017/invoice_db
REDIS_URL=redis://127.0.0.1:6379 | JWT_SECRET=your_jwt_key | SMTP_HOST=smtp.gmail.com

# 3. Execution Commands
npm run dev          # Development Mode
npm run start:prod   # Production PM2 Cluster
```

API REFERENCE ENDPOINTS

Method	Endpoint	Description	Auth
POST	/api/auth/register	Register user or business account	✗ Public
POST	/api/auth/login	Authenticate user and return JWT token	✗ Public
POST	/api/invoices	Create invoice record and compute financial totals	✓ Bearer JWT
GET	/api/invoices	Retrieve all historical invoices (Cached via Redis)	✓ Bearer JWT
GET	/api/invoices/:id/download	Generate and stream downloadable PDF invoice	✓ Bearer JWT
POST	/api/invoices/:id/email	Automatically email invoice PDF to client inbox	✓ Bearer JWT
DELETE	/api/invoices/:id	Delete invoice record and clear Redis cache	✓ Bearer JWT
GET	/health	System diagnostics (MongoDB, Redis, Memory usage)	✗ Public

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